

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks:30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I Sanjanaa Sree.P (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sanj

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(93)

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Ques 1). Job Interview Outcome

Given :

(*) If an applicant passes the interview, then the applicant is hired.

(*) Farhan passed the interview.

Goal :

Prove that Farhan is hired using the Resolution algorithm.

Step 1 : Represent the statements in Propositional logic.

Let, P = Farhan passed the interview

H = Farhan is hired

Knowledge Base :

1. $P \rightarrow H$

2. P

Goal : H

Step 2 : Convert statements into CNF

The implication must be removed using the rule :

$$A \rightarrow B \equiv \neg A \vee B$$

$P \rightarrow H$ becomes : $\neg P \vee H$

The CNF clauses are :

$$C1 = \neg P \vee H$$

$$C2 = P$$

Negated Goal : Goal = H Negated Goal = $\neg H$ $C3 = \neg H$

Step 3: Apply Resolution Algorithm

Resolution Step 1:

Resolve C_1 and C_2 : $C_1 = \neg P \vee H$ $C_2 = P$

Since P and $\neg P$ are complementary literals, they cancel each other.

Resulting Clause: $C_4 = H$

Resolution Step 2:

Resolve C_4 and C_3 : $C_4 = H$ $C_3 = \neg H$

Again, H and $\neg H$ are complementary literals.

Result = Empty Clause.

Therefore, Farhan is hired.

Ques 2). Attendance Credit

Given:

(i) If a student attends all classes, then the student gets attendance credit.

(ii) Heena attended all classes.

Goal:

Prove that Heena gets attendance credit using the

Resolution Algorithm.

Step 4 : Resolution Procedure

First Resolution : Resolve C_1 and C_2 .

$$C_1 = \neg A \vee C$$

$$C_2 = A$$

The literals A and $\neg A$ cancel each other.

$$\text{Result : } C_4 = C$$

Second Resolution : Resolve C_4 and C_3 .

$$C_4 = C$$

$$C_3 = \neg C$$

Result : Empty Clause.

$\therefore$ Meena gets attendance credit.

Ques 3). Car Rental Eligibility :

(i) Propositional Logic Representation :

L = Suresh has a valid license

R = Suresh can rent a car.

Knowledge Base : $L \rightarrow R$

L

Goal : R

(ii) convert into CNF and negate goal :

Implication Removal :

$$1. L \rightarrow R = \neg L \vee R$$

Step 1: Representation in Propositional Logic

Let, A = Meena attended all classes.

C = Meena gets attendance credit.

1. $A \rightarrow C$

2. A

Goal: C

Step 2: Conversion into Conjunctive Normal Form (CNF):

$$A \rightarrow B \equiv \neg A \vee B$$

The Statement: $A \rightarrow C$ becomes $\neg A \vee C$.

The CNF clauses are $C1 = \neg A \vee C$

$$C2 = A$$

Step 3: Negation of the goal.

The goal is C

Negating the goal: $\neg C$

Add this clause to the clause set.

$$C3 = \neg C$$

Complete Clause Set:

$$C1 = \neg A \vee C$$

$$C2 = A$$

$$C3 = \neg C$$

clauses :

$$C1 = \neg L \vee R$$

$$C2 = L$$

$$C3 = \neg R$$

(iii) Resolution Steps :

Resolve $C1$ and $C2$:

$$C1 = \neg L \vee R$$

$$C2 = L$$

Result :

$$C4 : R$$

Resolve $C4$ and $C3$:

$$C4 : R$$

$$C3 : \neg R$$

Result : Empty clause ; Surash Can Rent a Car.

Ques 4). Sports Achievement Award :

(i) Propositional logic Representation :

S = Divya scored above 90%.

M = Divya wins a gold medal.

Knowledge Base :

$$S \rightarrow M$$

S

Goal : M

(ii) Convert into CNF and Negate Goal:

Implication Removal: $S \rightarrow G = \neg S \vee G$

Clauses: $C1: \neg S \vee G$

$C2: S$

$C3: \neg G$

(iii) Resolution Steps:

$C1 = \neg S \vee G$

$C2: S$

Result: $C4: G$

Resolve $C4$ and $C3$: $C4: G$ $C3: \neg G$

Empty clause $\Rightarrow$ Darya wins a gold medal.

Ques 9). Access Control System

(i) convert to CNF: (ii) Resolution Algorithm:

$P \rightarrow A = \neg P \vee A$

Step 1: Resolve $C1$ and $C3$

$A \rightarrow G = \neg A \vee G$

$C1: \neg P \vee A$

$C3: P$

$C1: \neg P \vee A$

$C5: A$

$C2: \neg A \vee G$

$C2: \neg A \vee G$

$C3: P$

$C5: A$

$C4: \neg G$

$C6: G$

$C6: \neg G$

Result: Empty clause

The User is granted Success.